



JUKEBOB

Technical Architecture Documentation

SYSTEM DESIGN

Version 1.0 • December 2025



OVERVIEW

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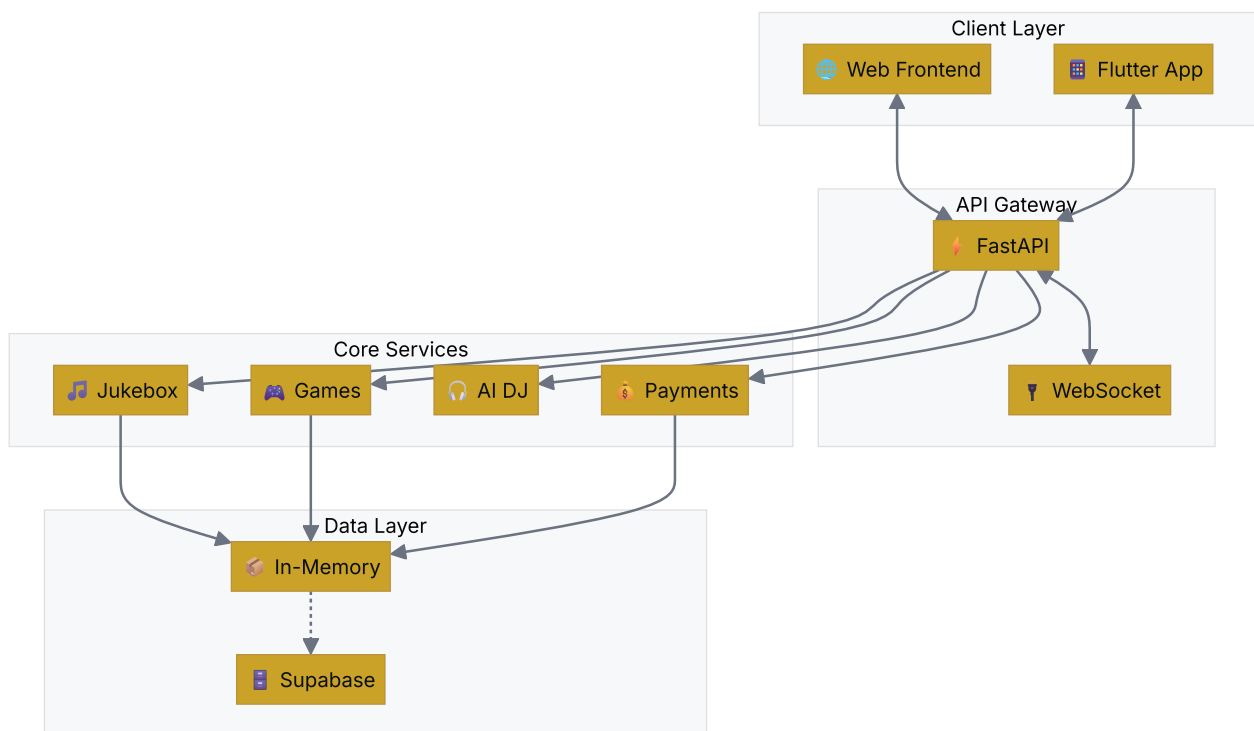
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ARCHITECTURE

01 System Architecture

Full-stack overview showing client applications, API gateway, core services, and data layer.

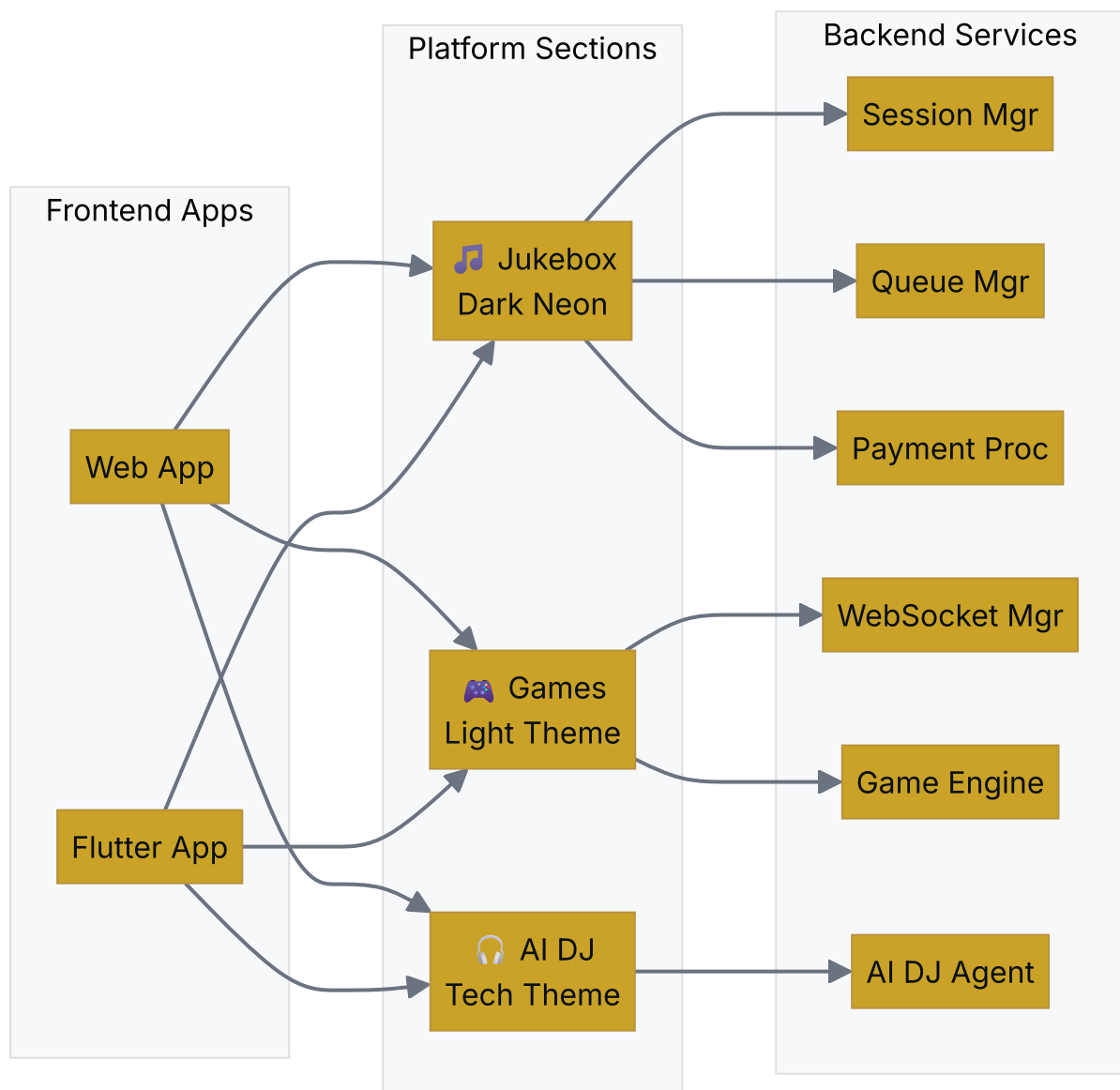




ARCHITECTURE

02 Component Diagram

Platform sections with dedicated themes and backend service connections.

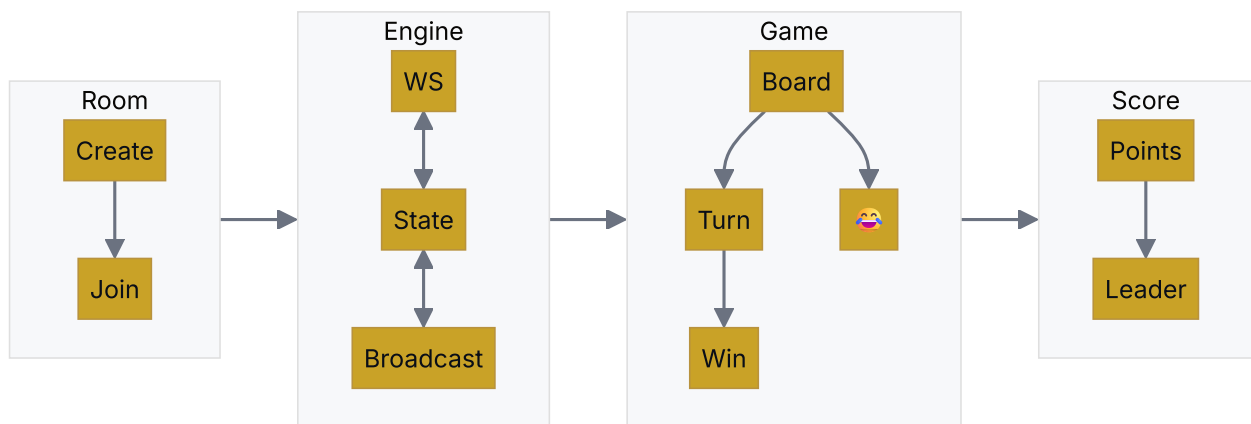




MODULES

03 Games Module

Real-time game engine with WebSocket sync and emoji reactions.

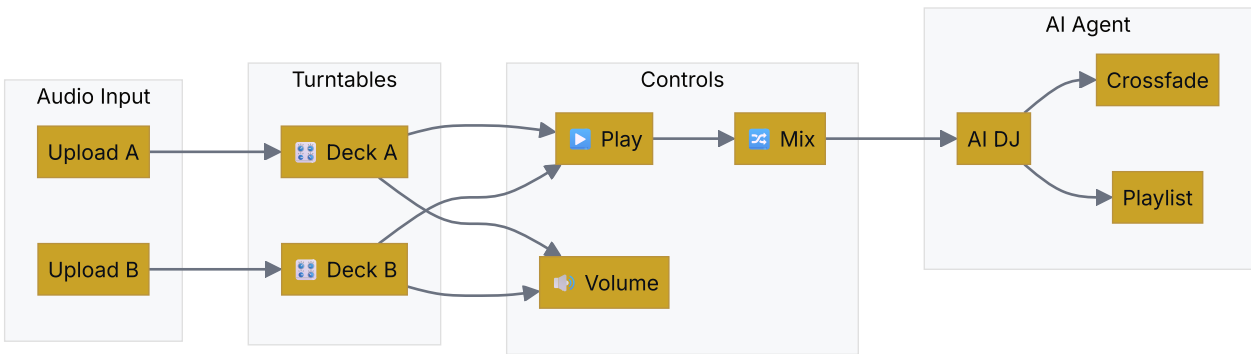




MODULES

04 AI DJ Module

Dual turntable system with AI-powered crossfading and mood analysis.

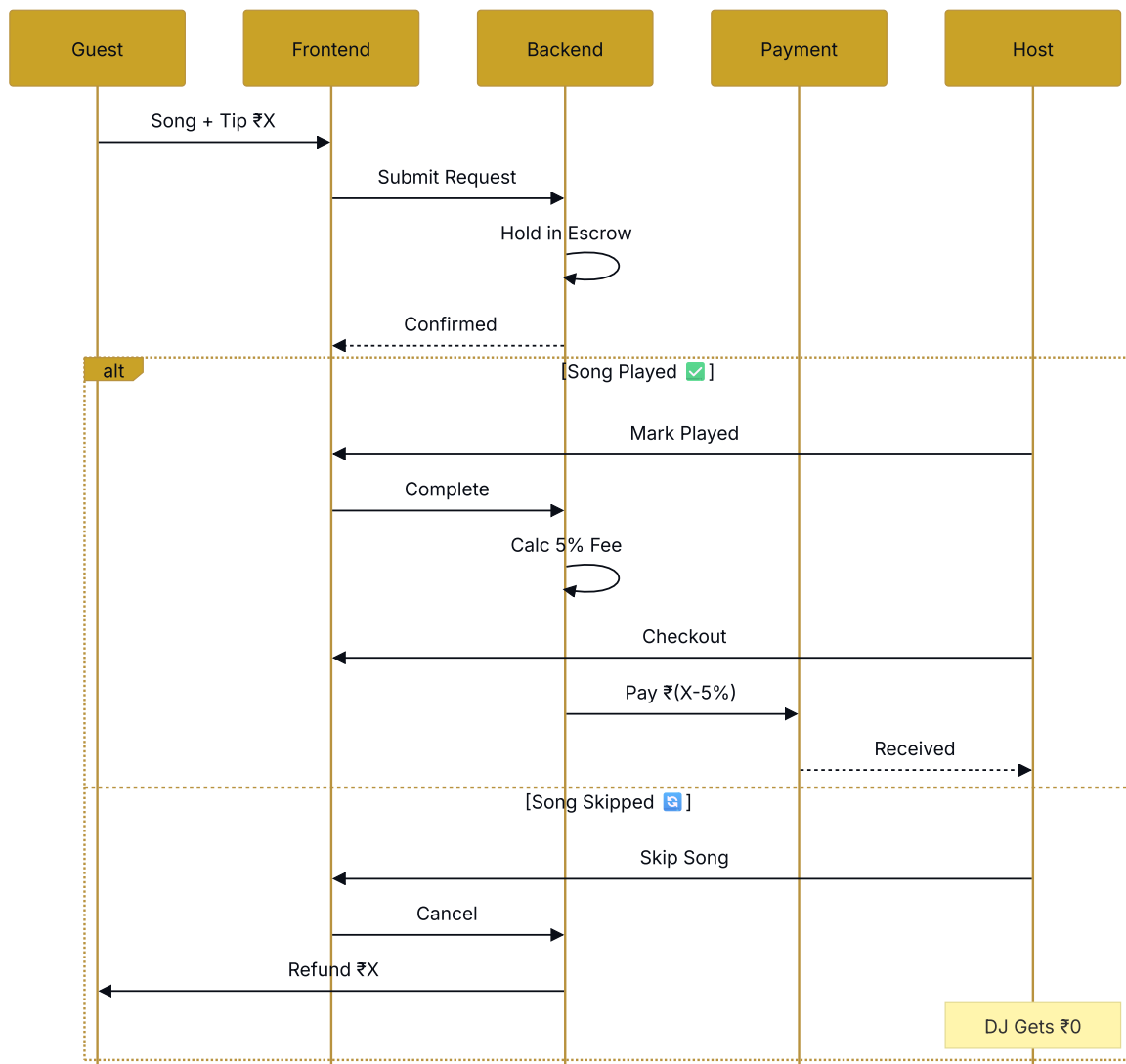




FLOWS

05 Payment Flow

Transaction flow with escrow protection. Tips refunded if song not played.

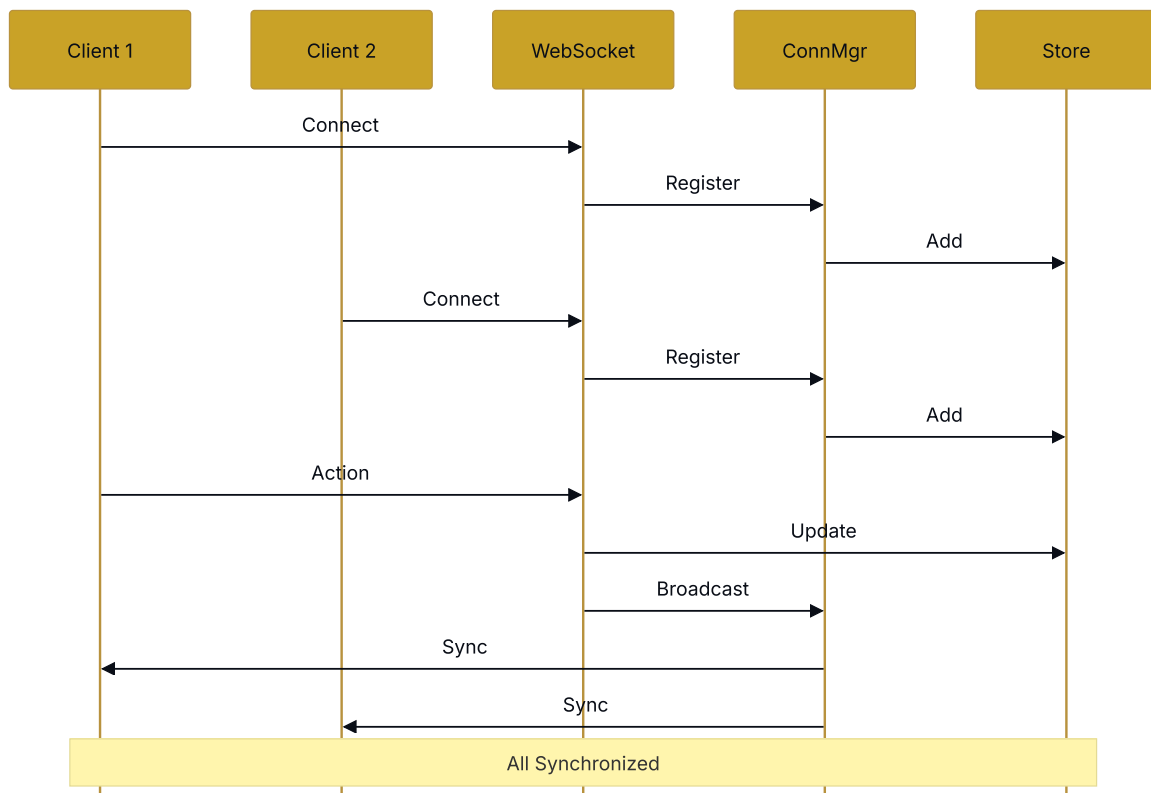




FLOWS

06 WebSocket Flow

Real-time synchronization keeping all connected clients in sync.



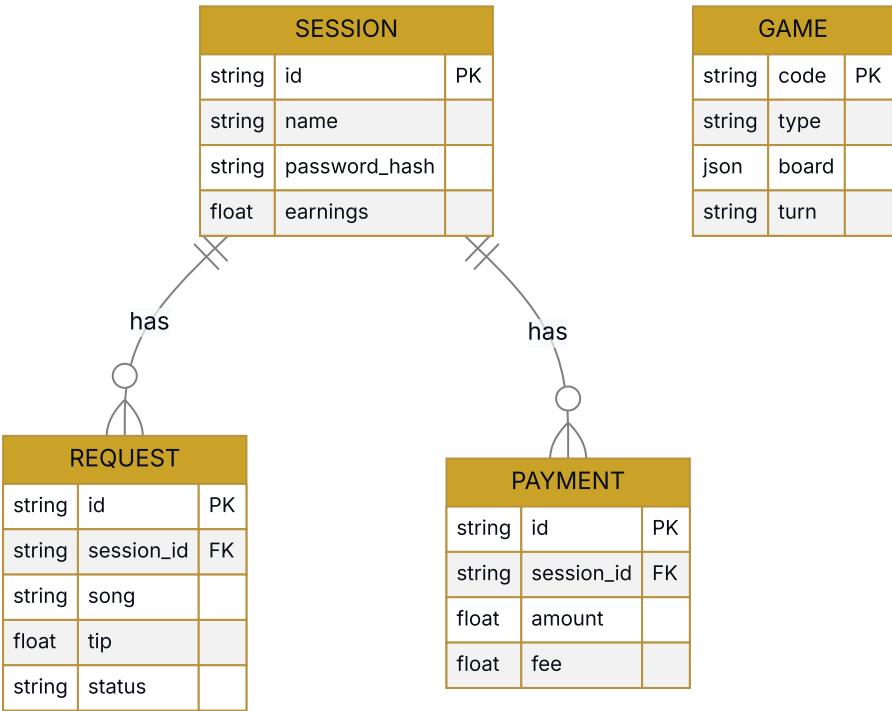


DATA

07

Data Model

Entity relationships for Sessions, Requests, Games, and Payments.





API

08

API Endpoints

REST API endpoints organized by service category.

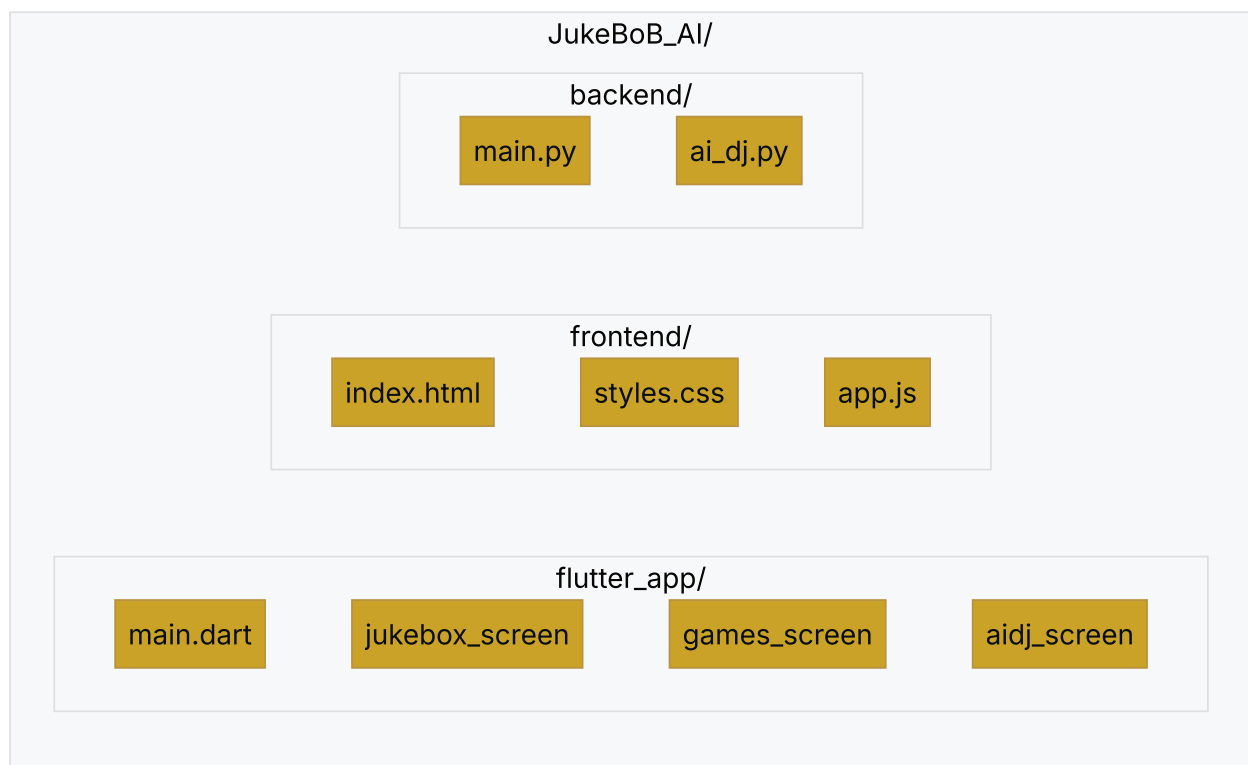
AI/OL APIs		Payment APIs		Game APIs			Request APIs			Session APIs		
POST /api/v1/ai/ol	GET /api/v1/ai/ol	POST /api/v1/payment/schedule	GET /api/v1/payment/revenue	POST /api/v1/games/create	POST /api/v1/games/delete	POST /api/v1/games/insert	POST /api/v1/request/create	POST /api/v1/request/delete	GET /api/v1/request/info	POST /api/v1/session/create	POST /api/v1/session/delete	GET /api/v1/session/info



STRUCTURE

09 Project Structure

File organization across backend, web frontend, and Flutter app.



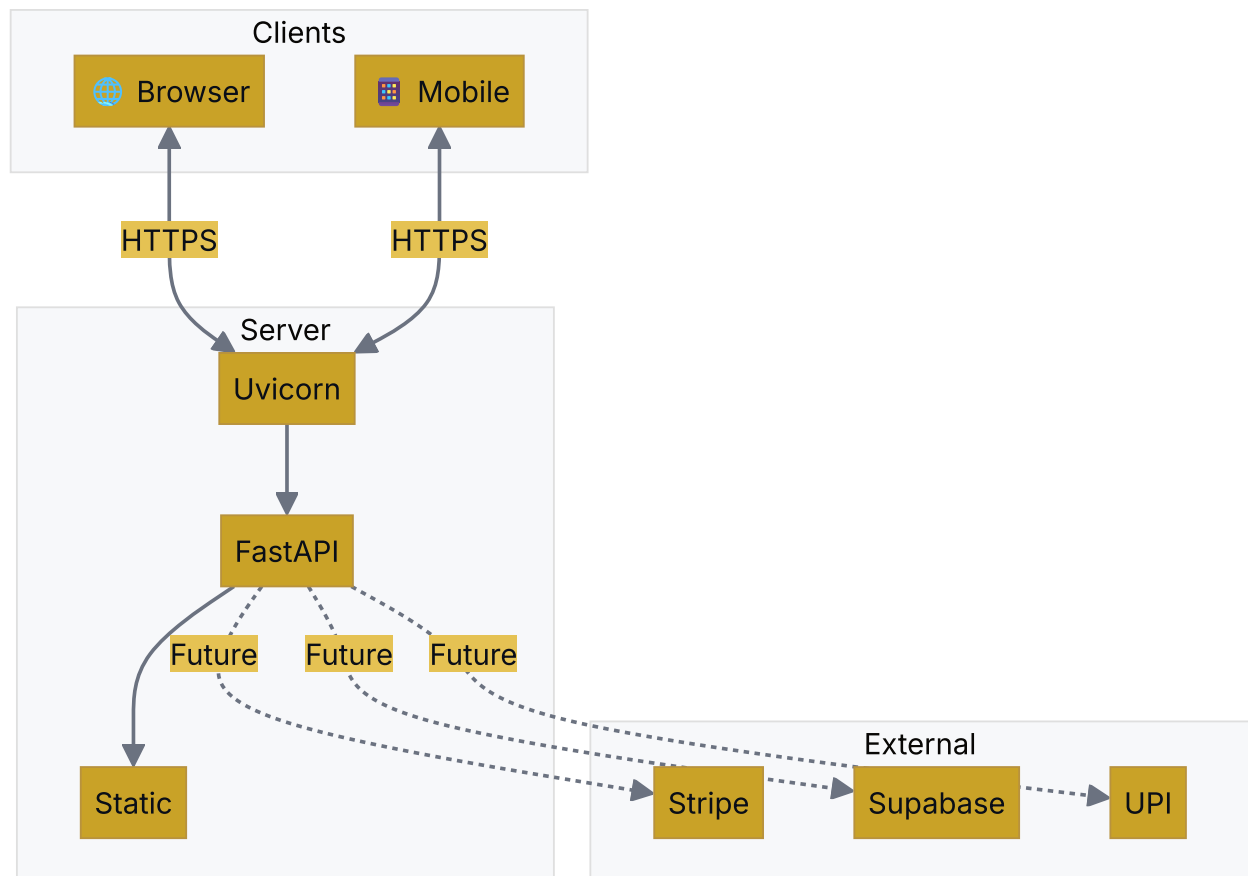


INFRASTRUCTURE

10

Deployment

Production deployment with external service integrations.

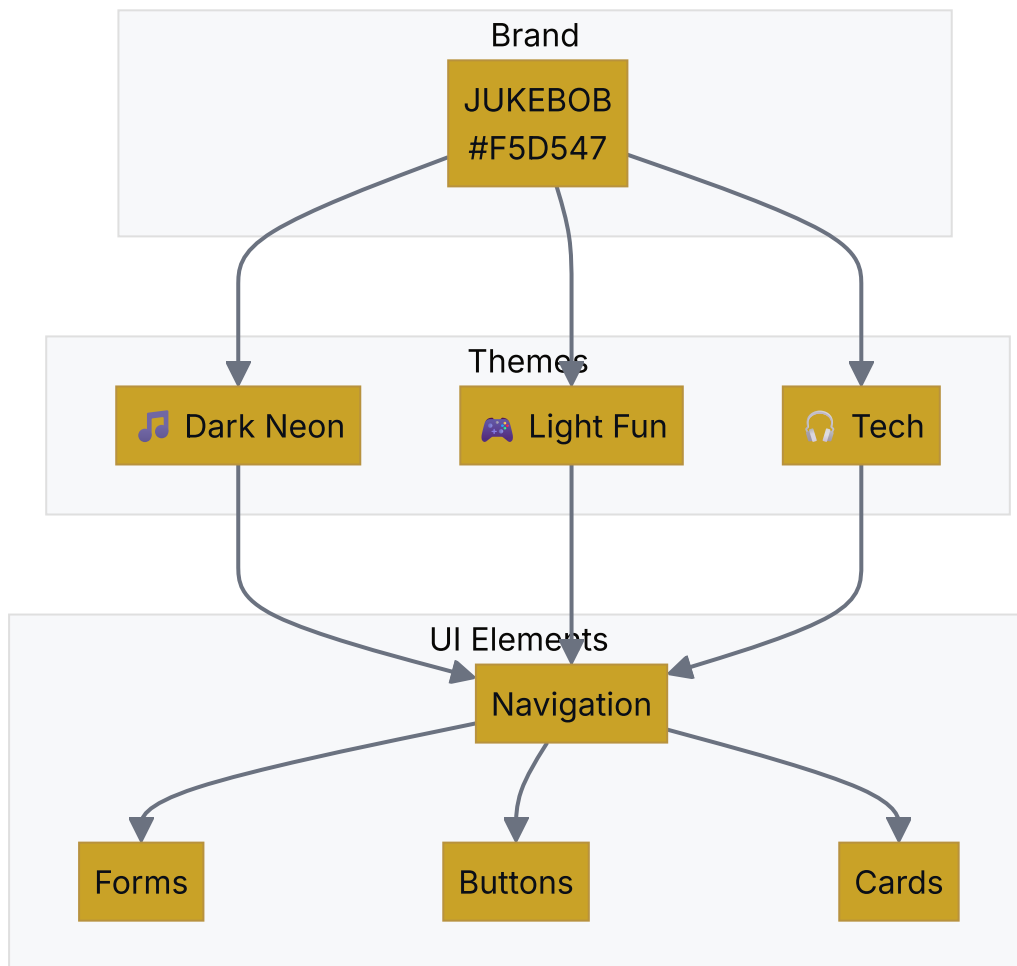




DESIGN

11 UI Architecture

Visual design system with brand identity and themes.

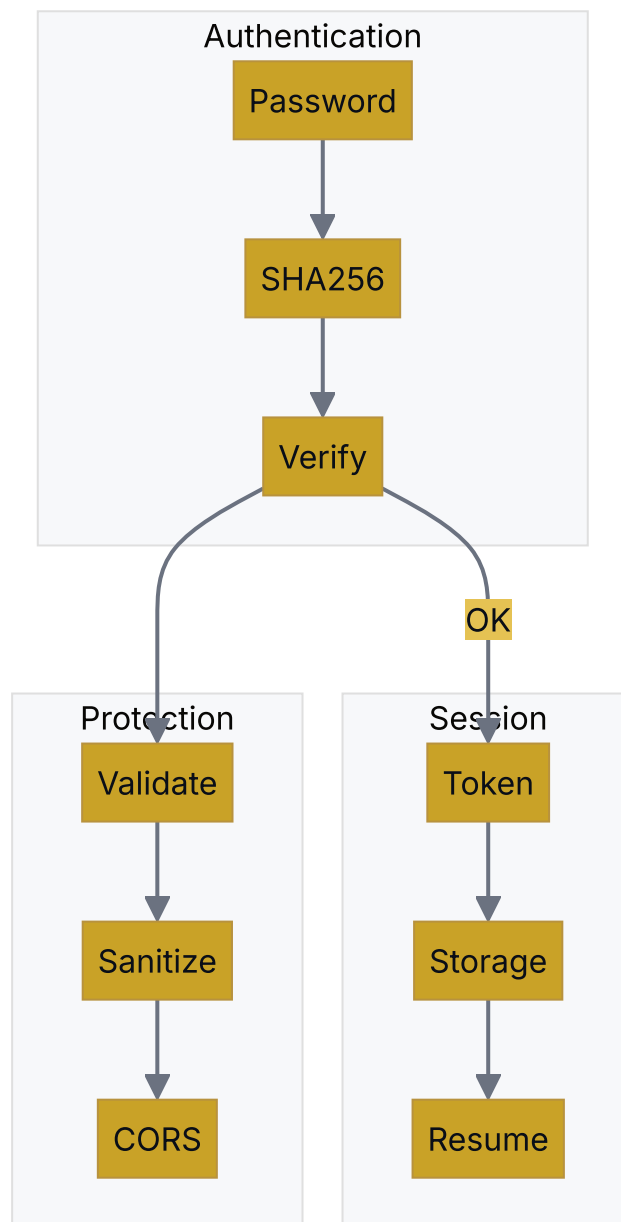




SECURITY

12 Security Architecture

Authentication, session security, and data protection.





NAVIGATION

13 State Management

User navigation flow through all platform sections.

